

YANAN HE

📍 New Haven, CT 🌐 <https://echoo113.github.io>

*I study why AI **works** when it works, why it **fails** when it doesn't, and how understanding those differences can help us **build better models**.*

EDUCATION

Yale University, New Haven, CT

Aug 2026 – May 2028

M.S. in Computer Science (*Fully Funded*)

Purdue University, West Lafayette, IN

Aug 2023 – May 2026

B.S. in Computer Science & Mathematics-Statistics

GPA: 3.96/4.0 | *Graduated with Distinction*

RESEARCH EXPERIENCE

NLP Lab, Yale University

New Haven, CT

Research Assistant | Advisor: Arman Cohan

Apr 2026 – Present

- Interpretability & Post-Training – investigating the mechanisms underlying **subliminal learning** and using these insights to improve how language models learn during post-training.
- LLM Evaluation, Alignment, and Long-Horizon Agents [1]

Graph Lab, Stony Brook University

Stony Brook, NY

Research Assistant | Advisor: Tengfei Ma

May 2025 – Apr 2026

- Time-Series Applications – JEPA-style architectures [2] and causal multi-scale modeling [3]
- LLM Reasoning in Geometric Space – hyperbolic latent navigation [4]

SELECTED PUBLICATIONS

1. Time to REFLECT: Can We Trust LLM Judges for Evidence-based Research Agents? * Equal contribution.
Leyao Wang*, **Yanan He***, Peng Chen, Asaf Yehudai, Yixin Liu, Rex Ying, Michal Shmueli-Scheuer, Arman Cohan
arXiv:2605.19196.
2. SC-JEPA: Stabilizing Latent Predictive Learning for Time-Series Anomaly Prediction
Yanan He, Yunshi Wen, Xin Wang, Tengfei Ma
SDM 2026; Forecast@ICML26.
3. CAAD: Causality-Aware Multivariate Time Series Anomaly Detection via Multi-Scale Alignment and Structural Causal Consistency
Xin Wang, Yunshi Wen, **Yanan He**, Haotian Xu, Youlan Zhao, Michel Ferreira Cardia Haddad, Tengfei Ma
SIGKDD 2026 Research Track.
4. HyperGuide: Hyperbolic Guidance for Efficient Multi-Step Reasoning in Large Language Models
Yuyu Liu, Haotian Xu, **Yanan He**, Sarang Rajendra Patil, Mengjia Xu, Tengfei Ma
arXiv:2605.24140.

INTERESTS & SKILLS

- **Programming Languages:** Python, Java, C/C++, Assembly, R, SQL, MATLAB, JavaScript, HTML/CSS
- **Languages:** English (Professional), Chinese (Native)